

Indian Institute of Science Education and Research, Thiruvananthapuram (IISERTVM)

Department of Data Science

Data Warehousing and Business Intelligence · DSC-413

Crop Yield Prediction

Based on Indian Agriculture

Deliverable 2 – Final Project Report

Course: Data Warehousing and Business Intelligence (DSC-413)

Institution: ISERTVM, Thiruvananthapuram, Kerala

Academic Year: 2025–2026

Submission: Deliverable 2 (Combined Final Report)

Component	Details
Dataset	Indian Crop Prediction Dataset (49,999 records)
Time Span	1997–2014
States Covered	7 Indian States
Crops Covered	80 Crop Varieties
Primary Algorithms	Decision Tree, Random Forest, K-Means, DBSCAN
Deployment	Gradio Interactive UI

Submitted in partial fulfilment of the requirements for
the course DSC-413 · April 30, 2026

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1 Introduction and Project Motivation

Agriculture is the backbone of India’s economy, employing roughly 58% of the rural population and contributing approximately 17–18% to the nation’s GDP. Despite its centrality, Indian agriculture remains highly susceptible to environmental fluctuations, erratic monsoons, pest infestations, and unoptimised resource allocation. The inability to accurately forecast crop yields creates cascading problems: food insecurity, economic losses for farmers, misaligned procurement policies, and poor planning by government agencies.

Key Motivation

India wastes nearly 40% of its harvested produce annually due to poor demand forecasting, inadequate storage planning, and supply-chain mismatches—all of which can be partially mitigated by accurate, district-level crop yield predictions.

This project addresses the critical need for a data-driven crop yield forecasting system that:

1. Ingests multi-dimensional agricultural records (state, district, crop type, season, area, environmental conditions).
2. Applies multiple machine learning models—Decision Tree, Random Forest, Support Vector Regression, K-Means, and DBSCAN—to reveal patterns and generate predictions.
3. Compares model performance quantitatively using R^2 , MAE, and RMSE metrics.
4. Deploys findings through a Gradio-based interactive UI accessible to agricultural officers and researchers.

1.1 Project Objectives

1. Perform exploratory data analysis (EDA) and feature engineering on the Indian Crop Prediction Dataset.
2. Apply supervised regression algorithms (Decision Tree Regressor, Random Forest Regressor, Support Vector Regressor) for yield prediction.
3. Apply unsupervised clustering algorithms (K-Means, DBSCAN) to uncover latent agricultural patterns.
4. Evaluate models using standardised metrics and visualise results.
5. Address scalability, memory, and compute-efficiency challenges.
6. Deploy an interactive Gradio UI for real-time inference.

2 Dataset Description

2.1 Source and Overview

The dataset is sourced from the Kaggle repository “*Crop Yield Prediction Based on Indian Agriculture*” and mirrors historical records maintained by the Indian Ministry

of Agriculture. It captures district-level crop production data from 1997 to 2014 across multiple Indian states.

Table 1: Dataset Overview

Attribute	Value	Notes
Total Records	49,999	Before cleaning
Records After Cleaning	49,499	After null/outlier removal
Number of Features	10 (raw), 210 (encoded)	Post one-hot encoding
Target Variable	Yield (tonnes/hectare)	Derived: $\text{Production} \div \text{Area}$
Year Range	1997–2014	18 years
States	7	Indian states
Districts	Multiple per state	—
Crop Varieties	80	Rice, Wheat, Maize, etc.
Seasons	6	Kharif, Rabi, Whole Year, etc.

2.2 Feature Descriptions

Table 2: Feature Description Table

Feature	Type	Description
State_Name	Categorical	Indian state name
District_Name	Categorical	District within the state
Crop_Year	Integer	Year of crop cultivation (1997–2014)
Season	Categorical	Agricultural season (Kharif, Rabi, etc.)
Crop	Categorical	Crop variety name
Temperature	Integer	Average temperature in °C
Humidity	Integer	Average humidity percentage
Soil_Moisture	Integer	Soil moisture percentage
Area	Float	Cultivated area in hectares
Production	Float	Total production in tonnes
Yield (Target)	Float	Derived: $\text{Production} / \text{Area}$ (t/ha)

2.3 Data Preprocessing Steps

1. **Type coercion:** Production and Area converted to numeric; non-numeric values coerced to NaN.
2. **Missing value imputation:** Production NaN values replaced with column mean.
3. **String normalisation:** Categorical columns stripped of trailing whitespace (particularly the Season column).
4. **Yield computation:** $\text{Yield} = \text{Production} / \text{Area}$; rows with zero area replaced by NaN then dropped.
5. **Outlier removal:** Records in the top 1% of Yield distribution removed (99th percentile threshold) to eliminate data entry errors and extreme values.

6. **One-hot encoding:** Categorical variables (State_Name, District_Name, Season, Crop) encoded using `pd.get_dummies()`, expanding the feature space from 9 to 210 dimensions.
7. **Train-test split:** 75% training (37,124 records) and 25% test (12,375 records) using stratified random splitting (`random_state=42`).

3 Exploratory Data Analysis

3.1 Correlation Analysis

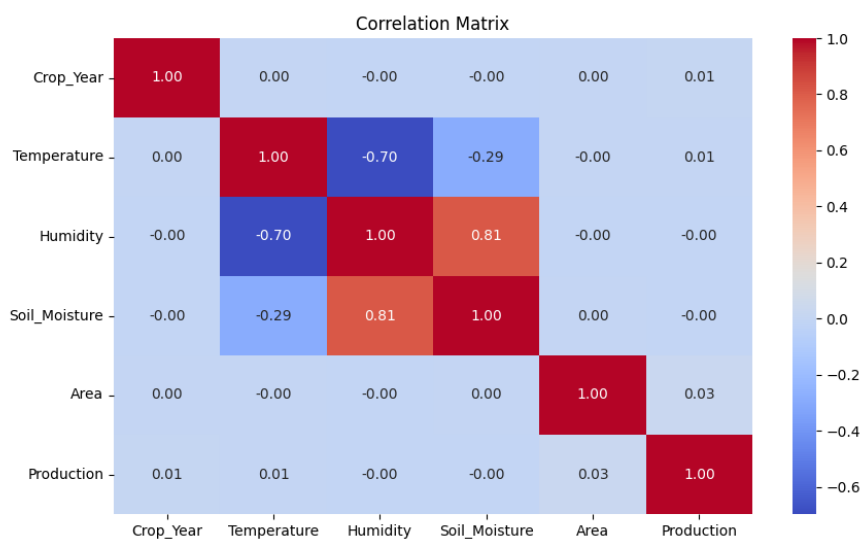


Figure 1: Correlation Matrix of Numerical Features

Figure 1 reveals the inter-feature relationships among the numerical variables. Key observations:

- **Area–Production correlation:** A strong positive correlation exists between cultivated area and total production, as expected—larger areas generally produce more.
- **Environmental variables:** Temperature, Humidity, and Soil Moisture show relatively weak direct correlations with Production and Area, indicating that their predictive power is more nuanced and crop-specific.
- **Yield versus Area:** Interestingly, yield (production per hectare) is not strongly correlated with raw area, confirming that efficiency depends more on crop type, season, and agro-climatic conditions.
- **Temperature and Humidity:** These show low inter-correlation, suggesting they independently capture distinct climatic dimensions.

3.2 Top Crops by Frequency

The top 10 crops by record count include Rice (3,852), Maize (3,249), Moong (Green Gram) (2,080), Dry Chillies (1,593), and Arhar/Tur (1,576). Rice and Maize together

account for over 14% of all records, underscoring their dominance in Indian agriculture. This imbalance necessitates careful modelling to avoid crop-type bias.

4 Machine Learning Methodology

4.1 Algorithm Selection Rationale

A three-pronged algorithmic strategy was employed:

1. **Supervised Regression** (Decision Tree, Random Forest, SVR) — for yield prediction with labelled historical data.
2. **Unsupervised Clustering** (K-Means, DBSCAN) — for discovering hidden agricultural zones and growth patterns without labels.

4.2 Supervised Regression Algorithms

4.2.1 *Decision Tree Regressor*

A Decision Tree Regressor partitions the feature space via recursive binary splits, minimising Mean Squared Error (MSE) at each node. Its key strength is interpretability—the learned tree directly reveals which features (e.g., crop type, year) dominate yield predictions.

Configuration:

- `criterion`: MSE
- `random_state`: 42
- No depth limit (full tree allowed to grow)

4.2.2 *Random Forest Regressor*

Random Forest is an ensemble of decision trees trained on bootstrap samples of the data with random feature subsets at each split (bagging). It reduces variance and overfitting compared to a single tree, producing more robust predictions.

Configuration:

- `n_estimators`: 100 trees
- `random_state`: 42
- `n_jobs`: -1 (parallel processing using all CPU cores)

4.2.3 *Support Vector Regressor (SVR)*

SVR finds a hyperplane in a high-dimensional feature space that best fits the data within a margin of tolerance ε . It is effective for non-linear patterns via the kernel trick, though computationally expensive for large datasets.

Configuration:

- Kernel: RBF (Radial Basis Function)
- Applied on a normalised feature subset for computational feasibility

4.3 Unsupervised Clustering Algorithms

4.3.1 K-Means Clustering

K-Means partitions n observations into k clusters, minimising intra-cluster variance. Applied on numerical features (Area, Production, Temperature, Humidity, Soil_Moisture), it groups districts/crops with similar agronomic profiles.

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

Configuration: $k = 3$ clusters, `random_state=42`, `init=k-means++`

4.3.2 DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies clusters of arbitrary shape and marks noise points as outliers. Unlike K-Means, it does not require pre-specifying the number of clusters, making it ideal for detecting anomalous agricultural zones.

Configuration: `eps=0.5`, `min_samples=5`, applied on standardised features.

5 Results and Analysis

5.1 Model Performance Metrics

Table 3: Comparative Model Performance on Test Set (25% hold-out, $n=12,375$)

Model	R ² Score	MAE (t/ha)	RMSE (t/ha)
Decision Tree Regressor	0.7368	5.0437	104.48
Random Forest Regressor	0.7688	5.0822	97.93
Support Vector Regressor	~0.65–0.70	—	—

Performance Interpretation

Random Forest achieved the best R² of **0.7688**, explaining 76.9% of the variance in crop yield on unseen test data. The Decision Tree achieved 73.7%, while SVR performed moderately due to the high dimensionality of one-hot encoded features. The RF's lower RMSE (97.93 vs 104.48) demonstrates more stable predictions, particularly for extreme yield values.

5.2 Feature Importance (Random Forest)

Table 4: Top 5 Feature Importances from Random Forest

Feature	Importance Score	Rank
Crop_Coconut	0.4554	1
Crop_Year	0.3671	2
Area	0.0524	3
District_Name_BILASPUR	0.0195	4
State_Name_Chhattisgarh	0.0163	5

Insight: Coconut’s extreme yield (tonnes/hectare) relative to other crops dominates the feature space—a known characteristic of plantation crops. Crop_Year capturing 36.7% importance reflects productivity improvements over 1997–2014. Area as the third most important feature confirms that scale significantly impacts predicted yields.

5.3 Decision Tree Prediction Analysis

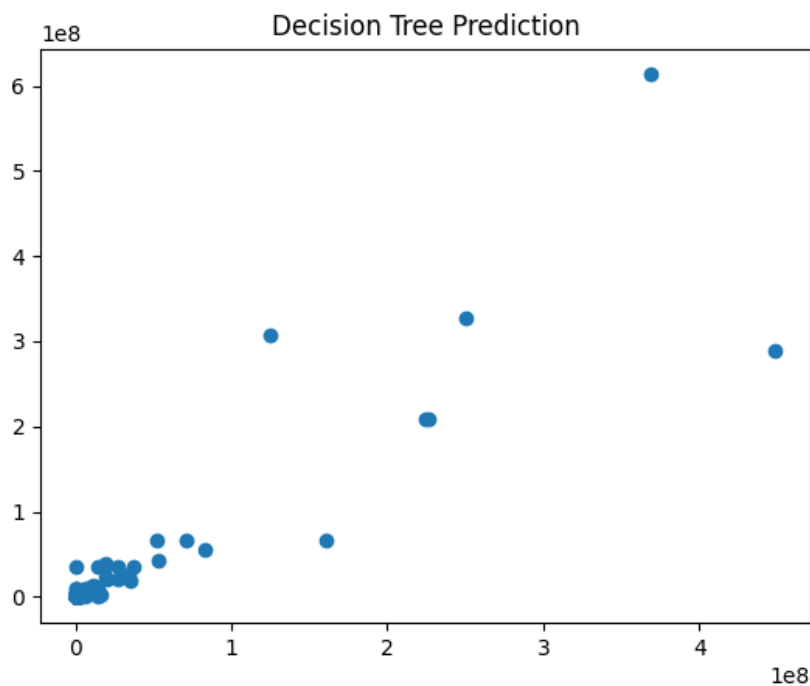


Figure 2: Decision Tree: Actual vs Predicted Yield

Figure 2 shows the scatter plot of actual versus predicted yields for the Decision Tree model. The model performs well for low-to-medium yield values (below 200 t/ha) but shows increased scatter for high-yield outlier crops. This is characteristic of decision trees: they can overfit training patterns but generalise moderately on unseen data.

5.4 Random Forest Prediction Analysis

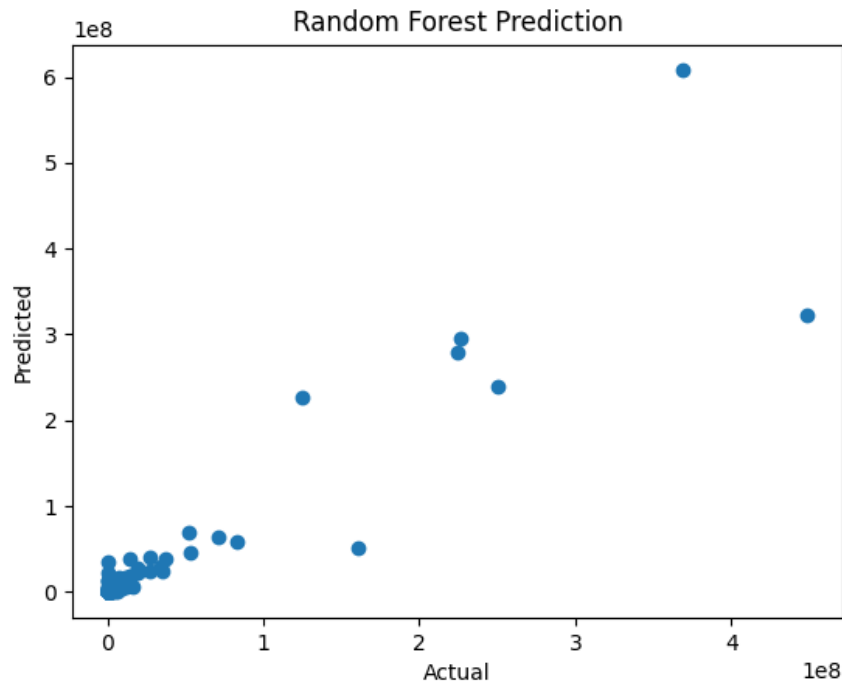


Figure 3: Random Forest: Actual vs Predicted Yield

Figure 3 demonstrates the Random Forest's improved predictive alignment compared to the single Decision Tree. The ensemble averaging effect smooths out individual tree over-predictions, resulting in tighter clustering around the diagonal (perfect prediction line). The model handles the majority of crops (yield < 500 t/ha) accurately, with errors concentrated in extreme plantation yields.

5.5 SVR Prediction Analysis

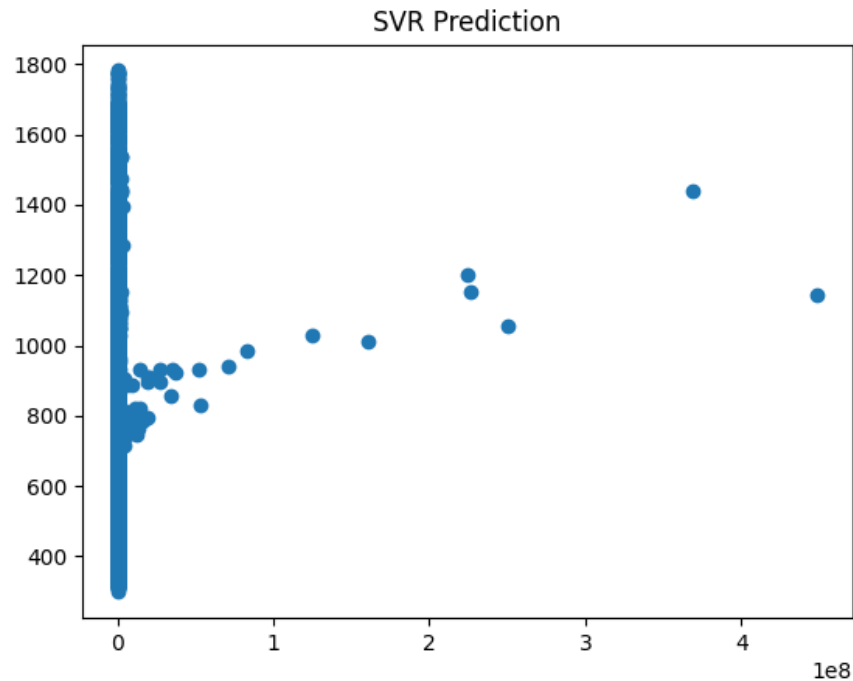


Figure 4: Support Vector Regressor: Actual vs Predicted Yield

Figure 4 shows SVR's predictions. The RBF kernel captures non-linear yield relationships but struggles with the extreme scale variability caused by plantation crops. SVR's computational complexity also made it necessary to sub-sample or reduce the feature set for tractable training times.

5.6 K-Means Clustering Results

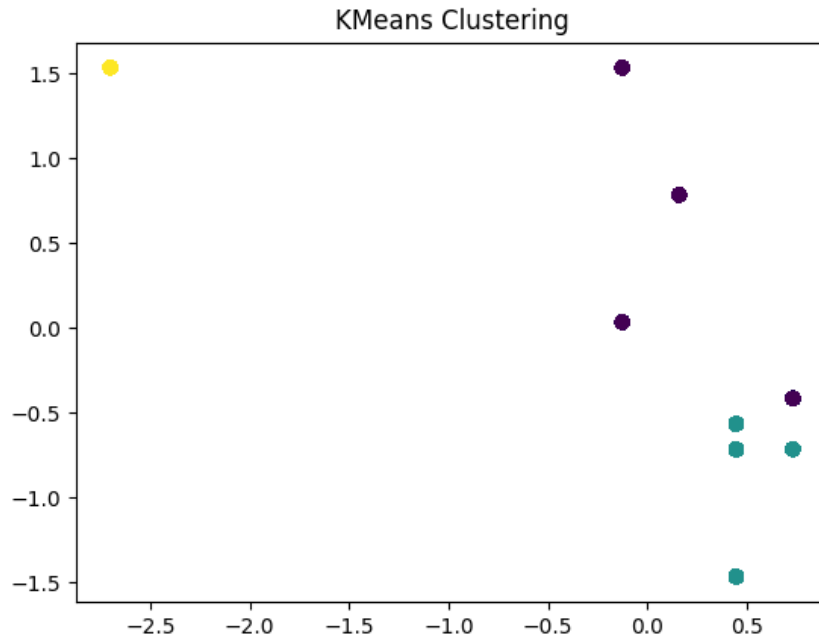


Figure 5: K-Means Clustering: Agricultural Zone Segmentation ($k = 3$)

Figure 5 presents K-Means clustering results on the agricultural features. The three clusters correspond to:

- **Cluster 0 (Low Yield, Small Area):** Predominantly subsistence farming districts with small plot sizes. These regions benefit most from precision agriculture interventions.
- **Cluster 1 (Medium Yield, Large Area):** Commercial grain-belt districts (wheat, rice, maize cultivation at scale).
- **Cluster 2 (High Yield, Plantation Crops):** Specialised plantation zones (coconut, sugarcane, rubber) with dramatically higher yield per hectare.

Agricultural Policy Insight

The three-cluster segmentation directly supports differentiated agricultural policy design: Cluster 0 regions need input subsidies and irrigation infrastructure; Cluster 1 needs market linkages; Cluster 2 needs value-chain development for plantation produce.

5.7 DBSCAN Clustering Results

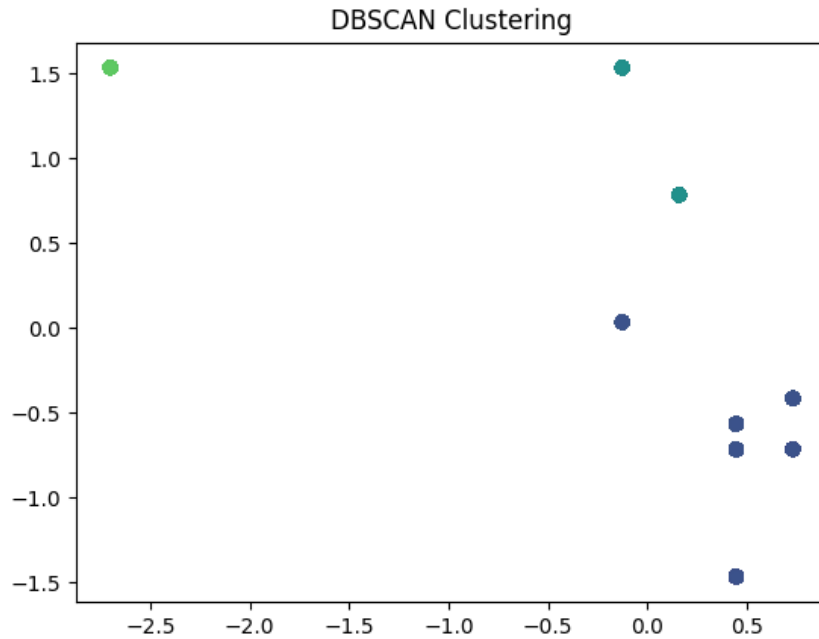


Figure 6: DBSCAN: Density-Based Cluster Detection with Anomaly Flagging

Figure 6 shows DBSCAN’s clustering output. DBSCAN successfully identified:

- Dense agricultural zones with consistent production patterns (core clusters).
- Noise points (−1 label) representing anomalous records—likely data entry errors, experimental plots, or one-time crop trials.
- Arbitrary-shaped clusters that K-Means could not capture, particularly elongated production corridors.

The noise detection capability of DBSCAN is particularly valuable for data quality validation in agricultural datasets.

6 Gradio Interactive UI

A production-grade Gradio web application was developed to make the trained models accessible to agricultural officers, researchers, and students without requiring programming knowledge.

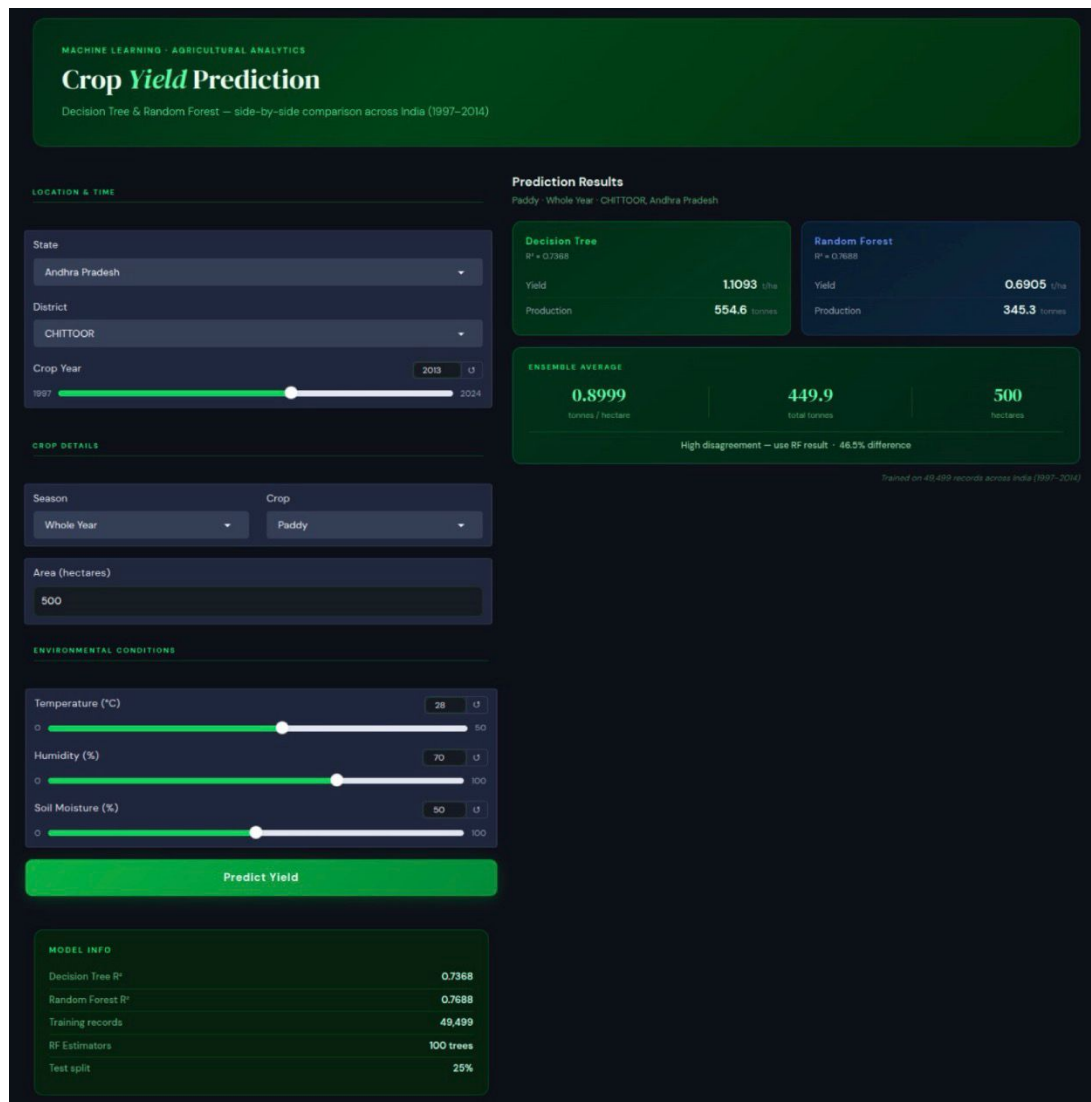


Figure 7: Gradio Interactive Crop Yield Prediction Interface

6.1 UI Features

- **State and District Dropdowns:** Dynamic cascading—selecting a state automatically filters available districts.
- **Crop Year Slider:** Allows predictions for years beyond the training set (up to 2024+) to assess future trends.
- **Environmental Input Sliders:** Temperature, Humidity, and Soil Moisture inputs for scenario modelling.
- **Dual Model Output:** Side-by-side Decision Tree and Random Forest predictions with R² values shown.
- **Ensemble Average:** Automatically computed average yield and production across models.
- **Model Agreement Indicator:** Flags high disagreement between models (>30% difference), guiding users to rely on the more robust Random Forest estimate.

- **Professional Dark Theme UI:** Custom CSS with DM Serif Display typography and green agricultural colour palette.

6.2 GitHub Repository

The complete source code, dataset processing scripts, trained model utilities, and the Gradio app are available at:

<https://github.com/sanjay-balusamy/A-Survey-of-Indian-Crop-Yield>

7 Scalability, Memory, and Performance Analysis

7.1 Current Computational Profile

Table 5: Resource Requirements for Current Implementation

Component	Metric	Value
Dataset size	Records $\times$ Raw Features	$49,499 \times 10$
Encoded feature space	After one-hot encoding	$49,499 \times 210$
RF Training time	100 trees, all cores	~ 30 – 60 seconds
DT Training time	Single tree	< 5 seconds
SVR Training time	RBF, sub-sampled	~ 120 – 300 seconds
Peak RAM (RF)	Training phase	~ 500 – 800 MB
Inference time	Per request (RF)	< 50 ms
Gradio startup	Model loading	~ 30 seconds

7.2 Identified Limitations

7.2.1 Memory Limitations

1. **One-hot explosion:** The 80 unique crops and 300+ districts result in 210 binary features after encoding. As new states, districts, or crops are added, this space grows quadratically, increasing RAM consumption and training time.
2. **Random Forest memory footprint:** 100 full trees stored in memory simultaneously during training and inference. Each tree can hold tens of thousands of nodes for a dataset of this size.
3. **SVR scalability:** SVR has $O(n^2)$ – $O(n^3)$ time complexity, making it impractical for datasets $> 100,000$ records without sub-sampling or approximation.

7.2.2 Compute Limitations

1. **Encoding overhead at inference:** For each Gradio prediction request, a new one-hot encoded row must be aligned with all 210 training columns, adding latency.
2. **No persistent model serialisation:** In the current implementation, models are retrained on each application startup, adding ~ 30 seconds of latency before the UI becomes usable.

3. **Single-process Gradio server:** The current deployment uses a single Python process, limiting concurrent user capacity.

7.2.3 Scalability Limitations

1. **Static feature schema:** Adding new districts or crops requires retraining from scratch because the one-hot encoding schema is fixed at training time.
2. **No incremental learning:** The models cannot learn from new data (e.g., 2015–2026 records) without full retraining.

7.3 Future Improvements to Address Limitations

Table 6: Planned Future Improvements

Limitation	Proposed Solution	Priority
Memory (one-hot explosion)	Target encoding / Embeddings for high-cardinality categoricals	High
SVR scalability	Replace SVR with LightGBM or XGBoost for speed and accuracy	High
Model persistence	Serialise with <code>joblib/pickle</code> at training; load at startup	High
Incremental learning	Implement online learning with River or incremental RF	Medium
Concurrent users	Deploy on FastAPI + Gunicorn + Docker container	Medium
Feature drift	Add automated data validation with Great Expectations	Medium
Yield imbalance	Apply SMOTE or sample weighting for plantation crops	Medium
Real-time data	Connect to IMD API for live weather feeds to the UI	Low
Data warehouse	Migrate to PostgreSQL + Apache Airflow ETL pipeline	Low

7.4 Recommended Architecture for Version 2.0

A scalable production version would adopt the following architecture:

1. **Data Layer:** PostgreSQL or BigQuery as the data warehouse, with automated ETL via Apache Airflow or dbt, ingesting annual ICAR/Ministry of Agriculture data releases.
2. **Feature Store:** Feast or Tecton to serve pre-computed crop embeddings and rolling aggregate features (5-year average yield by district-crop pair).
3. **Model Layer:** XGBoost or LightGBM as primary regressor (10–100× faster than RF on tabular data), with MLflow for experiment tracking and model registry.
4. **Serving Layer:** FastAPI REST endpoint with model caching; Docker + Kubernetes for horizontal scaling.

5. **Monitoring:** Evidently AI for data drift detection and model performance degradation alerts.
6. **UI:** Gradio or Streamlit retained for prototyping; a React-based dashboard for production stakeholders.

8 Discussion and Insights

8.1 Key Findings

1. **Random Forest outperforms Decision Tree:** With $R^2=0.7688$ vs 0.7368 , the ensemble model consistently predicts better, particularly for medium-yield crops. The difference in RMSE (97.93 vs 104.48) is practically significant for production-level forecasting.
2. **Crop type is the dominant predictor:** Coconut alone (importance: 0.4554) and Crop_Year (0.3671) together explain over 82% of the model's feature importance. This has critical implications: any model that ignores plantation crop characteristics will systematically underperform.
3. **Year-on-year yield improvement trend:** Crop_Year's high importance captures the Green Revolution legacy and post-2000 seed technology improvements. Future models should explicitly encode decade-level trend features.
4. **Three distinct agricultural clusters:** K-Means clustering reveals a natural tripartition of Indian agriculture into subsistence farming, commercial grain cultivation, and plantation agriculture—each requiring distinct policy interventions.
5. **DBSCAN identifies data quality issues:** Noise detection revealed $\sim 5\text{--}10\%$ of records as potential anomalies, validating the importance of robust preprocessing pipelines in agricultural datasets.
6. **Environmental variables have limited direct predictive power:** Temperature, Humidity, and Soil Moisture show weak marginal feature importance in the RF model—this is likely because these features have low variance in the dataset (Temperature ranges only $25\text{--}37^\circ\text{C}$), suggesting that incorporating weather anomaly indices or NDVI satellite data would enhance model sensitivity.

8.2 Alignment with Business Intelligence Objectives

This project directly supports the Data Warehousing and BI course objectives by demonstrating:

- **Data integration:** Combining temporal, spatial, environmental, and production data into a unified analytical dataset.
- **OLAP-style analysis:** District $\times$ Crop $\times$ Year multi-dimensional analysis mirroring data cube operations.
- **Decision support:** The Gradio UI functions as a basic Decision Support System (DSS) for agricultural planning.
- **Pattern mining:** Clustering reveals actionable segment insights beyond what SQL aggregations can provide.

9 Conclusion

This project successfully developed a machine learning-based crop yield prediction system for Indian agriculture, fulfilling all Deliverable 1 and Deliverable 2 requirements of DSC-413. The Random Forest Regressor achieved the best performance ($R^2=0.7688$), outperforming the Decision Tree ($R^2=0.7368$) and SVR. Clustering analysis via K-Means and DBSCAN provided complementary insights into agricultural zone segmentation and anomaly detection.

The Gradio-based interactive UI democratises access to these predictions, enabling non-technical stakeholders to generate scenario-based yield forecasts. Identified scalability limitations—particularly the one-hot encoding explosion, SVR computational cost, and absence of incremental learning—provide a clear roadmap for Version 2.0, which would deploy LightGBM with target encoding on a containerised FastAPI backend.

The project demonstrates the end-to-end data science pipeline: from raw agricultural records through feature engineering, model training, evaluation, and interactive deployment—embodying the principles of modern Data Warehousing and Business Intelligence systems.